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Data Analytics

Assignment 4
Level: 4000

Understanding What Drives Spotify Premium Willingness

Abstract

Most businesses and online services rely on the ability to convert free-tier users into paid subscribers. Understanding specific user behaviors and demographic factors that predict this conversion is one of the most important things for a B2C company like Spotify. This study investigates the correlation between Spotify user behavior, such as listening frequency, app usage, and satisfaction with recommendation algorithms, and their willingness to purchase a Premium subscription. By employing exploratory data analysis, statistical hypothesis testing, and predictive modeling, this project aims to identify the strongest indicators of subscription willingness to inform data-driven product decisions.

Introduction

Exploring the features that drive users to come back and potentially pay for a subscription is important for successful consumer technology. Spotify is a great example of a product with a lot of diverse users in terms of age group, subscription and free users, use frequencies, and more. As someone who is interested in product management, UX research, and data analytics, I wanted this project to showcase the intersection of these three fields.

I am also an avid Spotify user. I rely on the platform daily for streaming music and listening to podcasts. On the app store, it is one of the most popular apps for streaming. I initially converted from a free user to a Premium subscriber because of my high daily usage, how long I have been using the app for, and my reliance on the platform's highly effective recommendation system for music discovery.

Based on my experiences, my initial hypothesis is that behavioral engagement metrics like high versatility of listening contexts and satisfaction with the recommendation system are the strongest predictors of a user's willingness to subscribe to Premium.

Subscription-based platforms like Spotify rely heavily on this conversion to remain profitable. The results from investigating this hypothesis can be directly applied to inform targeted marketing campaigns, optimize the free-tier user experience, and prioritize feature development to maximize conversion rates for Spotify or other similar applications.

Data Description

This analysis uses the "Spotify User Behavior Dataset," a free public dataset sourced from Kaggle. Having pre-determined a research interest related to Spotify and its users, I wanted to look for a dataset that captured diverse user interactions with the platform. The original dataset comprises 520 rows of anonymized, survey-based records that show valuable insights into behavior patterns of the user base.

During the initial data inspection, I ensured that the dataset had a mix of both free-tire and Premium subscribers so that the data is not too skewed, allowing comparative analysis.

The original dataset contained 20 distinct variables: age, gender, spotify_usage_period, spotify_listening_device, spotfiy_subscription_plan, premium_sub_willingness, preferred_premium_plan, preferred_listening_content, fav_music_genre, music_time_slot, music_influential_mood, music_lis_frequency, music_expl_method, music_recc_rating, pod_lis_frequency, fav_pod_genre, preferred_pod_format, pod_host_preference, preferred_pod_duration, and pod_variety_satisfaction. Feature selection was necessary to tightly focus the analysis. To answer the research question, I chose to focus on specific behavioral and demographic predictors like age, music_recc_rating (user satisfaction with the recommendation algorithm), music_lis_frequency (the versatility of a user's listening contexts), and spotify_usage_period (overall user longevity on the application). The primary target variable for this study is premium_sub_willingness.

Preliminary Analysis

Before conducting exploratory and statistical analyses, a preliminary assessment of the dataset was required to understand its structure and prepare the variables for predictive modeling. The first step in the preliminary analysis was to verify the completeness of the dataset. Using the colSums(is.na()) function, I checked for missing values across all columns to ensure that no records had to be deleted. Surprisingly, there were no missing values in this dataset.

A critical step in the data preparation phase was structurally re-ordering the ordinal variables. By default, R processes categorical variables alphabetically. For example, it placed "1 year to 2 years" before "6 months to 1 year". To ensure chronological accuracy in the subsequent visualizations and models, the Age and spotify_usage_period columns were explicitly converted into factored levels ordered sequentially from youngest to oldest, and shortest to longest usage, respectively.

A unique challenge discovered during the preliminary assessment was the music_lis_frequency column, which tracks the contexts in which users listen to music. This includes leisure time, workout, study time, etc. Most form answers had a combination of a variety of listening contexts. Because users could select multiple contexts, the raw data contained over 35 unique, comma-separated string combinations. Treating these as standard categorical variables would have resulted in fragmented and unusable data. To resolve this, I made a new quantitative feature called context_count. This uses the stringr package to count the delimiters within each response. I then generated a numeric value representing the versatility of a user's listening habits. See Figure 1 for the number of different listening contexts by count. To visualize the overall popularity of individual contexts, I parsed the delimited strings into individual observations, allowing for an accurate count of each specific listening environment (see Figure 2).

With the data cleaned and structurally formatted, cross-tabulations were generated to assess the baseline distributions of the user base. The demographic overview immediately highlighted that the user base is predominantly young adults, highly concentrated in the "20-35" age bracket (see Figure 3). Overall user satisfaction was also evaluated, showing a generally positive skew in how users rated the

recommendation system (see Figure 4). Finally, an initial assessment of app loyalty revealed a strong presence of long-term users, with the majority having used the platform for over two years (see Figure 5).

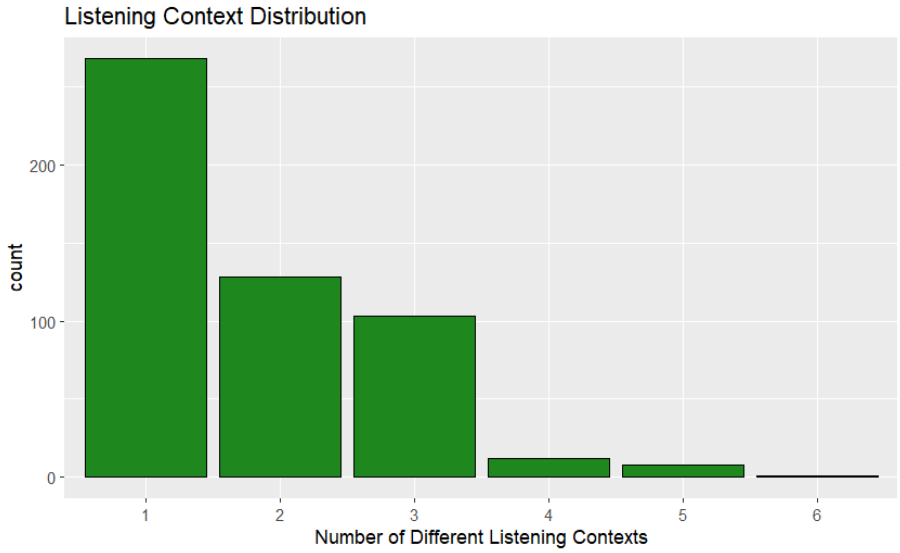

Figure 1: Number of Different Listening Contexts by Count

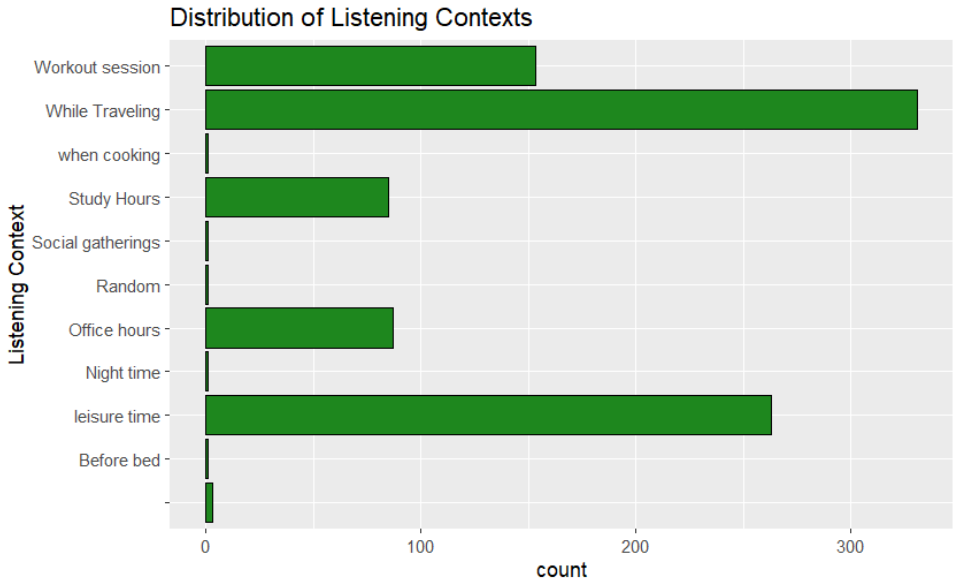

Figure 2: Distribution of Listening Contexts from the Survey

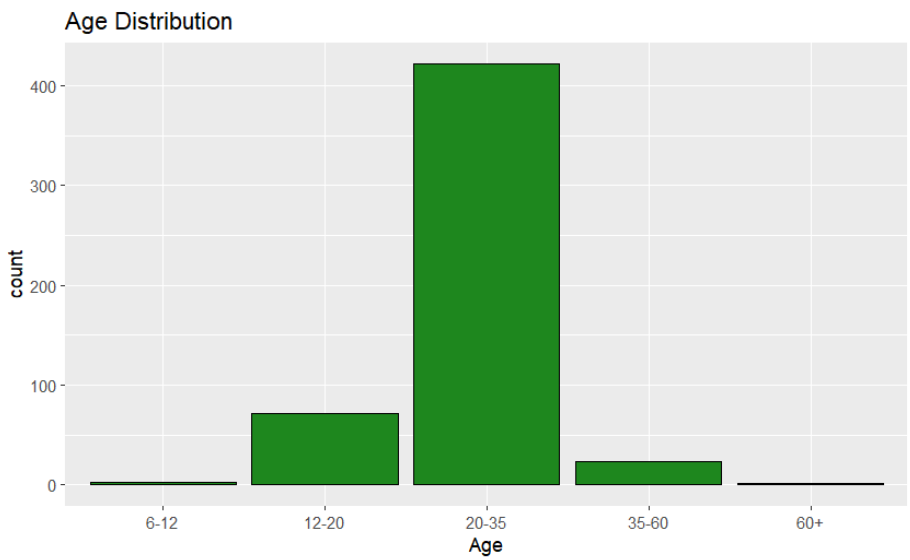

Figure 3: Histogram of Age Distribution

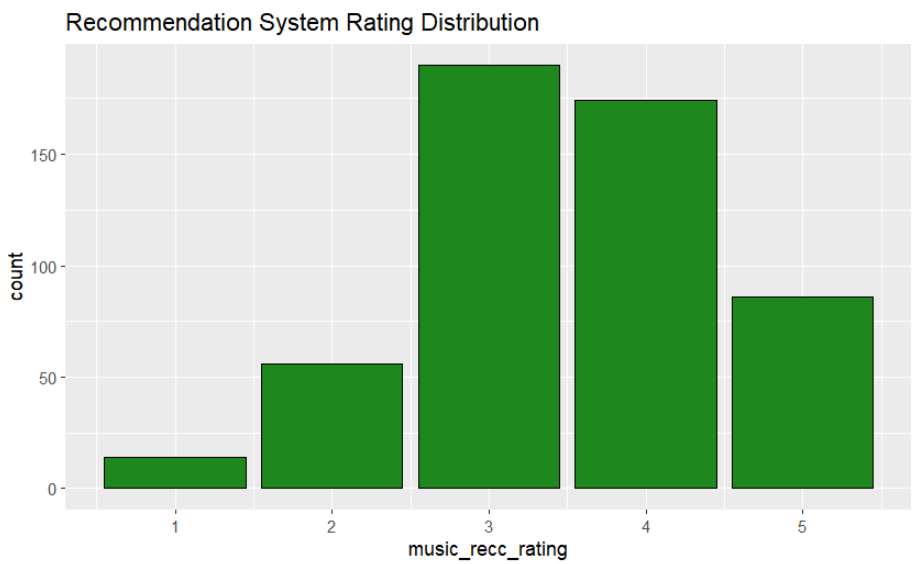

Figure 4: Distribution of Recommendation System Ratings

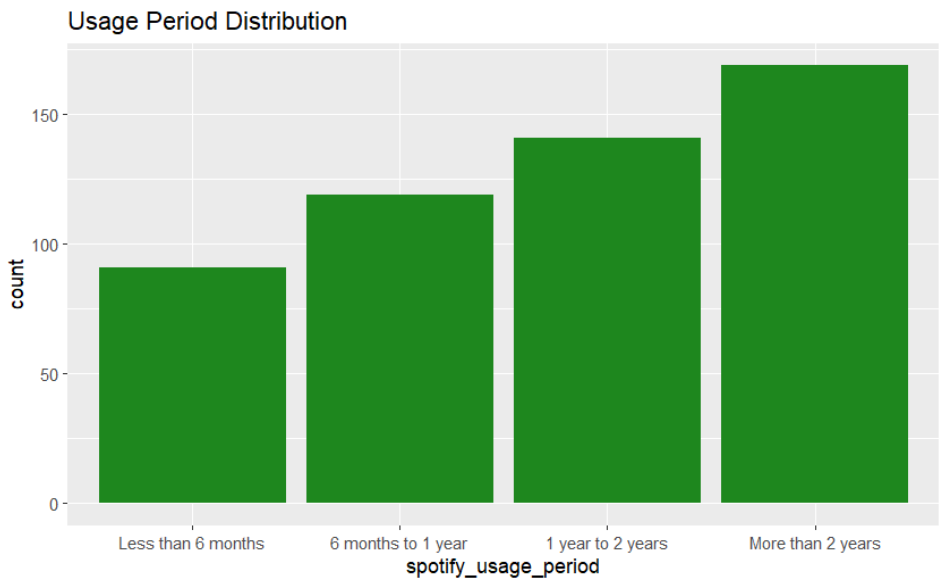

Figure 5: Distribution of Usage Periods

Exploratory Analysis

Following the preliminary data preparation, an in-depth exploratory data analysis was conducted to evaluate the distributions of the selected features. The goal was to detect underlying relationships with the target variable, premium_sub_willingness. I wanted to test the initial hypothesis: high behavioral engagement (longevity and a variety of listening contexts) and high satisfaction with the recommendation algorithm are the primary drivers of Premium conversion.

To begin exploring the hypothesis, bivariate visualizations were generated to compare the independent variables against subscription willingness. A grouped bar chart comparing music_recc_rating and premium_sub_willingness visually indicated a uniform distribution as seen in Figure 6. Most users rated the recommendation system highly with a lot of 3s, 4s, and 5s. However, the proportion of users willing to subscribe did not appear to increase sequentially with higher ratings. Users are perfectly happy getting great recommendations on the free tier, so Spotify needs to use other features to convince users to upgrade to premium.

Similarly, evaluating the context_count variable against subscription willingness revealed that users who listen to Spotify in many different environments are visually no more likely to subscribe than those who only listen in one or two contexts (see Figure 7). To round out the visual analysis, demographic and loyalty factors were examined. A grouped bar chart comparing Age against subscription willingness (Figure 8) showed that while the 20-35 demographic dominates the dataset, the proportion of users willing to subscribe remains relatively consistent across the available age brackets. Conversely, examining the usage period (Figure 9) provided the first visual indication of a trend, suggesting that users who have been on the platform for over a year represent a visually larger proportion of potential Premium subscribers. While app usage longevity (spotify_usage_period) showed visual promise as a dividing factor, statistical hypothesis testing was required to mathematically validate these visual interpretations.

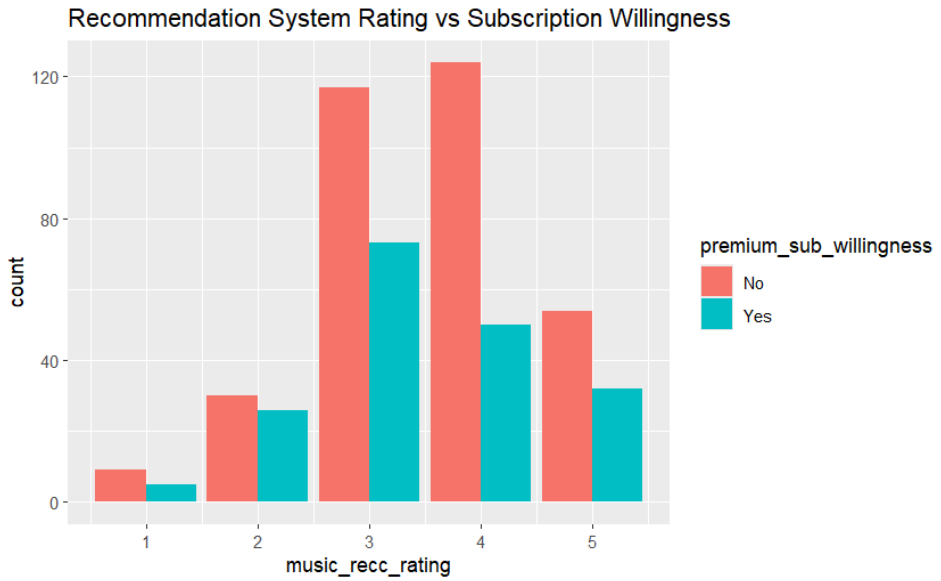

Figure 6: Recommendation System Rating vs Subscription Willingness

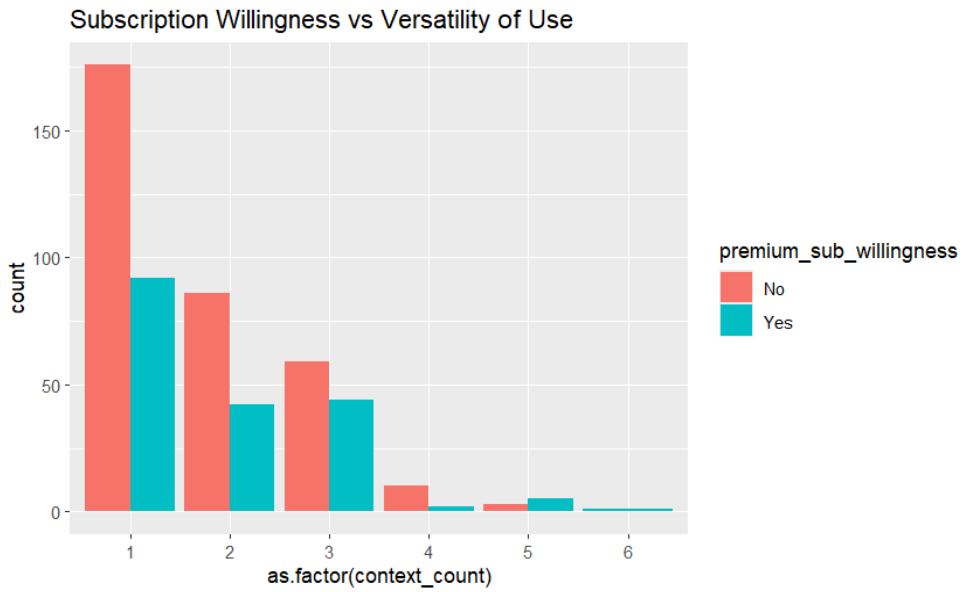

Figure 7: Subscription Willingness vs Versatility of Use

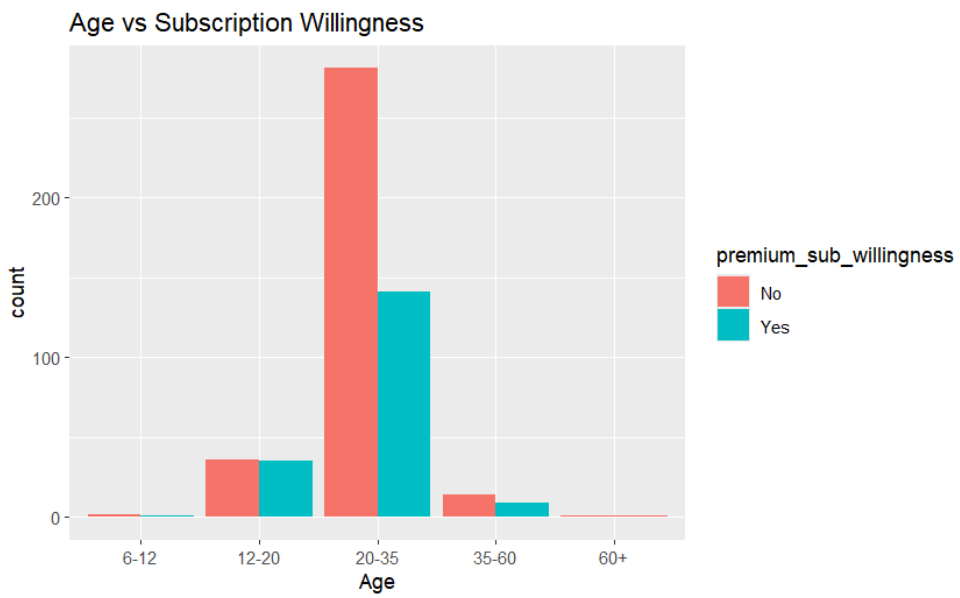

Figure 8: Age vs Subscription Willingness

Figure 9: Usage Period vs Subscription Willingness

Statistical Hypothesis Testing

To test the validity of the visual patterns, a series of statistical tests were performed to determine feature importance. For the continuous numerical variables, Welch Two Sample t-tests were utilized to compare the means of the "Yes" and "No" subscription willingness groups. For the categorical variables, Pearson's Chi-squared tests of independence were applied. A standard alpha level of 0.05 was used to determine statistical significance.

Recommendation System Rating (t-test): The null hypothesis is that there is no difference in the mean recommendation rating between users willing to subscribe and those who are not. The test resulted in a p-value of 0.1496 ($t = 1.444$, $df = 363.22$). Because the p-value is greater than 0.05, we fail to reject the null hypothesis. Surprisingly, the mean rating for non-subscribers is 3.55, which is slightly higher than for potential subscribers (3.42). This contradicts the initial hypothesis that algorithmic satisfaction drives monetization.

Listening Context Versatility (t-test): The null hypothesis is that there is no difference in the mean number of listening contexts between users willing to subscribe and those who are not. Testing the context_count variable gave a p-value of 0.1592 ($t = -1.411$, $df = 342.33$). Since the p-value is greater than 0.05, we fail to reject the null hypothesis again. This means that the number of environments in which a user listens is not a statistically significant predictor of Premium conversion.

Age Demographic (Chi-Square test): The null hypothesis is that there is no association between a user's age demographic and their willingness to subscribe. Testing the independence of Age and subscription willingness resulted in a p-value of 0.1184 ($X\text{-squared} = 7.3524$, $df = 4$). Demographic age is not a statistically significant driver of subscription intent in this dataset.

Spotify Usage Period (Chi-Square test): The null hypothesis is that there is no association between a user's overall usage period and their willingness to subscribe. The final test evaluated user longevity. This test yielded a p-value of 0.000228 ($X\text{-squared} = 19.38$, $df = 3$). We reject the null hypothesis and conclude that the length of time a user has been on the platform is strongly and mathematically associated with their willingness to upgrade to a Premium tier.

While the statistical findings are robust, several sources of potential bias and uncertainty must be acknowledged when interpreting these results.

Sources of Error, Uncertainty, and Bias

One important source of bias stems from the fact that this dataset is obtained from a survey. Since all variables are self-reported, responses may be influenced by recall bias or inconsistent interpretation of survey questions. For example, users may overestimate their listening frequency or rate the recommendation system more favorably due to familiarity with the platform rather than objective satisfaction. This can introduce a measurement error in the analysis. There can also be selection bias in how the data was collected. The dataset was taken from Kaggle, and the responses were most likely from voluntary survey participation. This may disproportionally capture responses from users who are more engaged online, and these individuals may already have stronger opinions about Spotify. Ultimately, this could distort relationships between engagement metrics and subscription willingness.

Additionally, the dataset is on the smaller side (520 observations) and not guaranteed to be representative of the global Spotify user base. The demographic distribution is skewed toward the 20-35 age group, which may limit the generalizability of the findings. If certain populations are underrepresented, the statistical relationships observed may not accurately reflect real-world behavior across all users.

Another source of uncertainty arises from the categorization of ordinal variables such as age and usage period. Although these were reordered correctly, the use of grouped ranges of ages reduces precision and may dismiss variability within the buckets. Two users at opposite ends of a range could exhibit very different behaviors but are treated identically in the analysis.

Finally, the statistical methods carry limitations. Because the dataset only contains 520 rows, the tests might miss smaller or weaker trends. Just because features like recommendation rating were "statistically insignificant" does not mean they have no impact in the real world. Additionally, it is crucial to remember that correlation does not equal causation. Even though there is a proven mathematical link between a long usage period and willingness to subscribe, simply using the app for a long time does not necessarily cause a user to pay.

These sources of bias and uncertainty suggest that while the findings provide meaningful insights, they may not be completely accurate and should be validated using larger, more representative datasets or experimental methods in future work.

Model Development and Application of Models

To predict a user's willingness to subscribe to Spotify Premium (premium_sub_willingness), I used two classification algorithms: Decision Tree and K-Nearest Neighbors (KNN) model. It was clear that this would be a classification task because the target variable, premium_sub_willingness, is a binary yes or no.

The Decision Tree is a supervised learning algorithm that partitions data by identifying the feature that provides the highest information gain at each node. It provides an interpretable, rule-based output that traces the reasoning behind each prediction. I chose this because of its high visual interpretability and its ability to natively handle complex, non-linear relationships without requiring feature scaling. The KNN algorithm classifies objects based on the closest training examples in the feature space. It requires no prior assumptions about the structure of the data. In this case, it classifies a new user by calculating their geometric distance to all its other users in the training set. It then assigns them the majority class of their nearest neighbors. I chose to use KNN to see if potential Premium subscribers naturally cluster together in a multi-dimensional feature space based on Euclidean distance. Both of these methods allow for cross-validation of findings and a robust evaluation of which features impact premium conversion.

Data Preprocessing

Before training the models, targeted preprocessing steps were applied to ensure algorithmic compatibility. Since the two models operate on different principles, they required separate versions of the dataset.

The Decision Tree uses the original categorical variables with their readable string labels which is essential for producing a visually interpretable tree plot with useful node annotations. I did not have to do any transformation to these variables for the Decision Tree.

On the other hand, the KNN model required a fully numerical feature space due to its reliance on Euclidean distance calculations. The ordinal categorical variables Age and spotify_usage_period were converted into integers that preserved their natural ordering. For example, spotify_usage_period was

mapped such that "Less than 6 months" = 1, "6 months to 1 year" = 2, "1 year to 2 years" = 3, and "More than 2 years" = 4. I converted the pre-ordered factor variables to numeric sequences using as.numeric(), using the chronological factor levels established during the preliminary data preparation phase.

Because KNN computes the geometric distance between data points across all features simultaneously, the scale of each variable has a direct impact on the model's behavior. In this dataset, music_recc_rating ranged from 1-5, context_count ranged from 1-6, and the encoded usage period ranged from 1-4. I standardized the numerical feature space by using standard z-score scaling (scale()) since they all had different scales. Without this normalization, variables with naturally larger numeric ranges would have disproportionately dominated the Euclidean distance calculations, heavily skewing the KNN model. This transformation centers each variable at a mean of zero and scales it to a standard deviation of one, ensuring every feature contributes proportionally to the distance calculation regardless of its original range.

Finally, the dataset was partitioned into an 80% training set (416 observations) and a 20% testing set (104 observations). A random seed (set.seed(123)) was fixed prior to partitioning to ensure the strict reproducibility of all splits and model evaluations. The same index vector was used to split both the numerical dataset (for KNN) and the raw categorical dataset (for the Decision Tree) so that both models were trained and evaluated on the same observations for a fair and direct comparison.

Decision Tree

The Decision Tree classifier was trained using R's rpart package with method = "class". The algorithm used the Gini impurity criterion to evaluate and select the best feature split at each node. Gini impurity measures how often a randomly chosen observation from a node would be incorrectly classified. At each level of the tree, rpart selects whichever feature and threshold minimizes the weighted Gini impurity across the resulting child nodes. The algorithm was allowed to grow naturally without manual pruning, relying on rpart's default parameters, which require a minimum of 20 observations to attempt a split and at least 7 observations in any terminal leaf node.

The tree was trained on the four selected features: Age, music_recc_rating, context_count, and spotify_usage_period. It uses the 80% training set with premium_sub_willingness as the target variable.

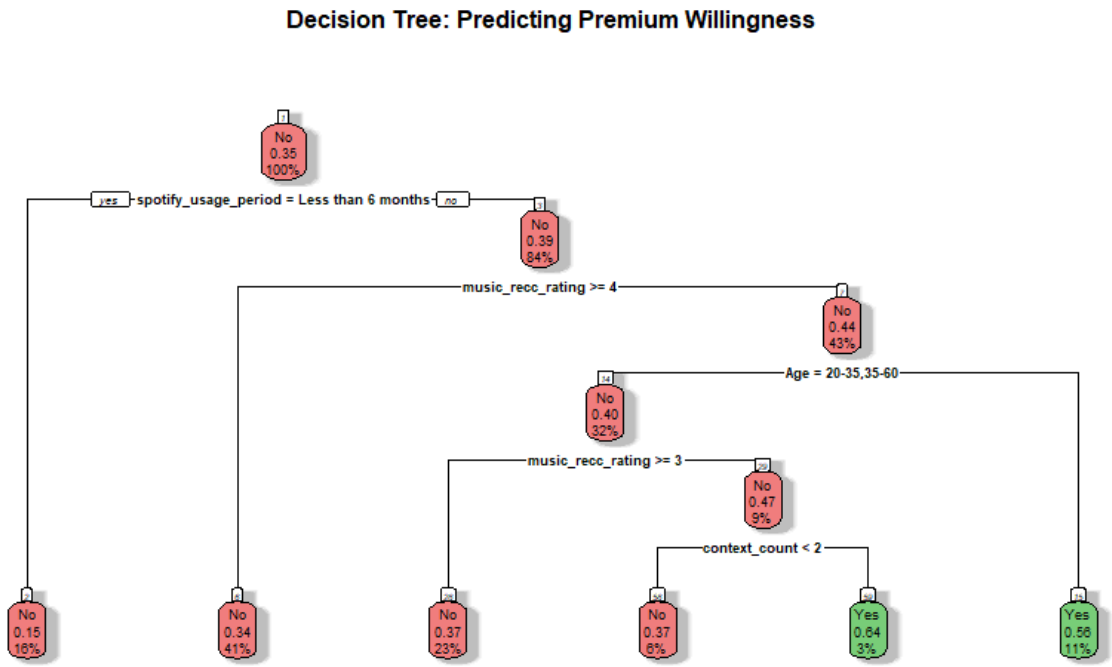

Figure 10: Decision Tree

The output of the Decision Tree (in Figure 10) validated the findings established during the Exploratory Data Analysis. The algorithm selected spotify_usage_period as the root node, confirming it as the strongest predictive feature. This is fully consistent with the chi-square test result, which identified usage period as the only statistically significant predictor ($p = 0.000228$). Users who had been on the platform for less than six months were separated into a left leaf node with only a 15% willingness rate. For longer-term users, the tree continued splitting on music_recc_rating and Age in the lower branches, with the only two "Yes"-majority leaf nodes appearing at the far right. This represents older, long-term users who rated the recommendation system highly and listened across multiple contexts.

When applied to the 20% validation testing data, the Decision Tree achieved an overall accuracy of 64.42%. However, calculating the confusion matrix revealed critical insights regarding the model's underlying behavior:

	Actual: No	**Actual: Yes**
Predicted: No	60	32
Predicted: Yes	5	7

Figure 11: Confusion Matrix for the Decision Tree Classifier.

The Decision Tree exhibited a highly "pessimistic" classification strategy. It proved exceptionally accurate at identifying true negatives, correctly predicting 60 users who ultimately would not subscribe. However, it identified only 7 true positives while generating 32 false negatives. Because the features lacked a strong distinguishing signal for subscribers, the algorithm learned a default rule: assume the user will remain on the free tier unless specific, narrow conditions are met.

K-Nearest Neighbors (KNN)

The KNN classifier was implemented using R's class package via the knn() function, applied to the standardized numerical feature set. Unlike the Decision Tree, KNN has no training phase. Instead, it stores the entire training set and makes predictions at runtime by finding the k training observations geometrically closest to each test point and assigning the majority class among those neighbors as the predicted label.

The distance between points is calculated using Euclidean distance across all four scaled features simultaneously. KNN treats every feature as a dimension in space. After scaling, each feature contributes equally.

Before running the final model, the feature matrix was separated from the target variable. The four predictor columns were isolated into train_x and test_x, while the target labels were stored separately in train_y and test_y.

Because KNN's behavior depends on the choice of k, it is important to select the best value. A small k makes the model sensitive to individual noisy points, while a large k smooths the decision boundary but risks oversimplifying it. Rather than choosing k arbitrarily, I developed a loop to run across all integer values from 1 to 20. For each value of k, the model was trained on the scaled training set and evaluated on the scaled test set, with the resulting accuracy recorded.

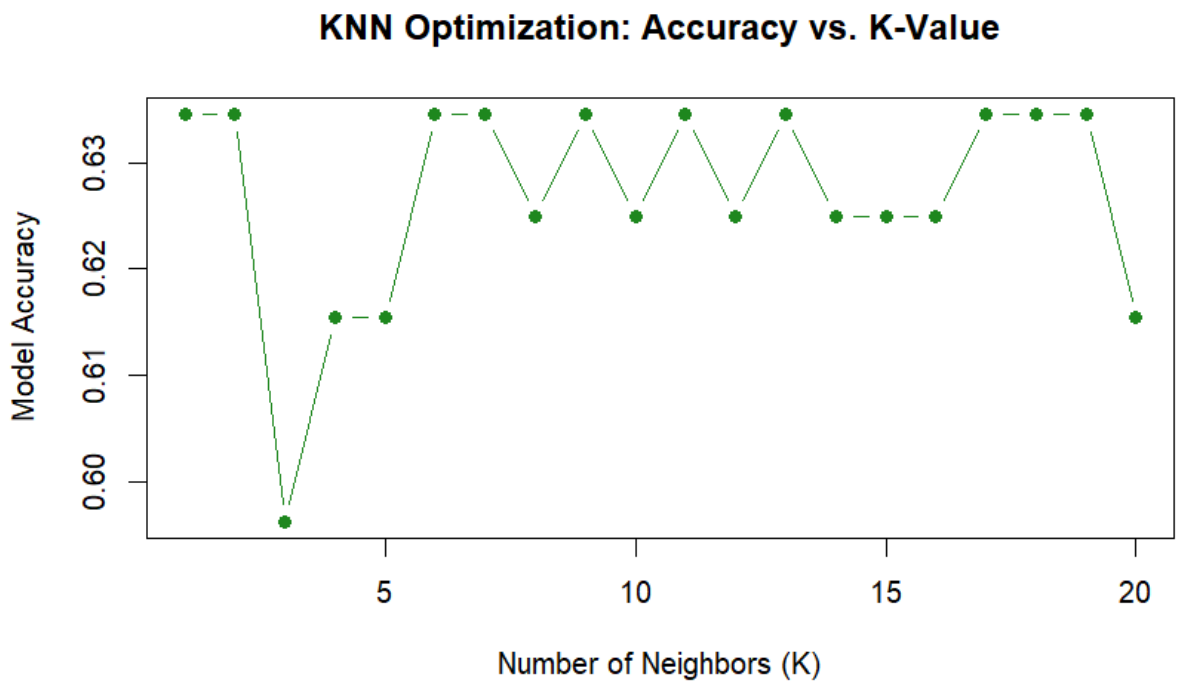

Figure 12: Hyperparameter tuning showing Model Accuracy across varying values of K

The optimization curve revealed significant volatility across k values, with accuracy ranging from approximately 59.6% at k = 3 to a peak of 65.38% at k = 1. The which.max() function automatically identified k = 1 as the optimal parameter and stored it as best_k, which was then passed directly into the final model.

However, k = 1 introduces an important theoretical concern. When k = 1, the model classifies every test point based on its single nearest neighbor in the training set. This makes it highly susceptible to noise and outliers. The optimization graph shows that several higher, more stable values such as k = 7 and k = 13 achieved nearly identical accuracy to k = 1. It may be preferred to use these values since they consult more neighbors, produce smoother decision boundaries, and generalize better to new data.

Using best_k = 1, the final KNN model was run and predictions were generated on the test set. The confusion matrix and accuracy were computed using the same approach as the Decision Tree for consistency.

	Actual: No	**Actual: Yes**
Predicted: No	55	26
Predicted: Yes	10	13

Figure 13: Confusion Matrix for the Optimized KNN Classifier (k=1).

The final KNN model achieved an accuracy of 65.38%. The confusion matrix shows that KNN captured 13 True Positives compared to the Decision Tree's 7, nearly doubling the recall rate for actual subscribers from 17.9% to 33.3%. KNN generated 10 False Positives versus the Decision Tree's 5, meaning it was slightly more aggressive in predicting "Yes." Still, 26 False Negatives remained, showing that both models ultimately share the same core limitation: the feature set does not provide a strong enough signal to reliably identify Premium subscribers.

Model Confidence and Comparative Analysis

Ultimately, the KNN model (65.38%) marginally outperformed the Decision Tree (64.42%). However, confidence in both models' ability to reliably classify future Spotify Premium subscribers remains limited.

The results show limitations within the dataset. Both models exhibited poor recall rates for actual subscribers because they were severely constrained by the input features. As proven by the earlier statistical hypothesis testing, features such as Age, Recommendation Rating, and Context Versatility do not mathematically differentiate a free user from a Premium user. Because the models were fed data that looks nearly identical for both classes, neither the rule-based Decision Tree nor the distance-based KNN could find a clean "signal" to isolate the Premium users. The rigorous testing and optimization of these models prove that to build a highly confident predictive tool for Spotify, the dataset must be expanded to include stronger indicators.

Conclusions and Discussion

The objective of this study was to identify the primary drivers of a user's willingness to subscribe to Spotify Premium. The results contradicted the initial hypothesis. The hypothesis was that high satisfaction with the recommendation algorithm and high versatility of listening contexts would strongly predict Premium conversion. However, statistical tests revealed that users on the free tier are equally satisfied with Spotify's music recommendations and listen in just as many diverse environments as those willing to pay.

Three of the four features (Recommendation Rating, Listening Context Versatility, and Age) showed no statistically significant relationship with subscription willingness. Only spotify_usage_period produced a significant result ($p = 0.000228$), revealing that how long a user has been on the platform is the single strongest indicator of Premium conversion. This directly contradicted the initial hypothesis and reframed the entire direction of the project. Users are not converted by how satisfied they are with recommendations or how versatile their listening habits are. They are converted by loyalty and time spent on the platform.

The predictive models reinforced this conclusion. Both the Decision Tree and KNN gave the same finding: the Decision Tree selected spotify_usage_period as its very first split, and both models had the same accuracy ceiling of 64-65%. This convergence across two different algorithms confirms that the limitation is in the data, not the modeling approach. Both models struggled most correctly identifying actual subscribers because the remaining features provided almost no discriminating signal between the two classes.

If I were to continue this project, the most impactful next step would be to get a dataset with stronger behavioral features. Other metrics like daily listening hours, whether a user has previously tried a free trial, or device type are more likely to separate free users from potential subscribers than survey-based satisfaction scores. Additionally, applying cross-validation instead of a single 80/20 split would produce more reliable accuracy estimates, and experimenting with another model like Random Forests can help quantify feature importance more precisely.

Ultimately, this project showed that data-driven product decisions require the right data first. The dataset placed a limit on what any model could achieve. In terms of Spotify, long-term users are the most convertible audience, and retention strategies that keep free users engaged over time are likely to be the most effective method for driving Premium subscriptions.

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